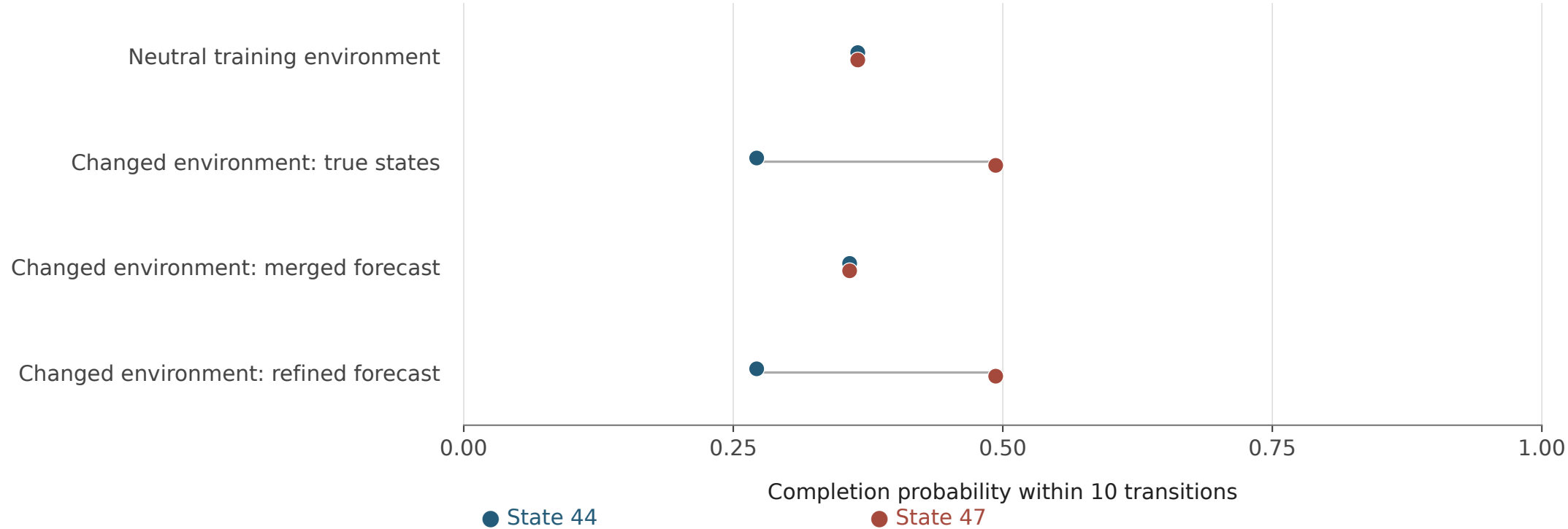
An intervention exposes a hidden state distinction

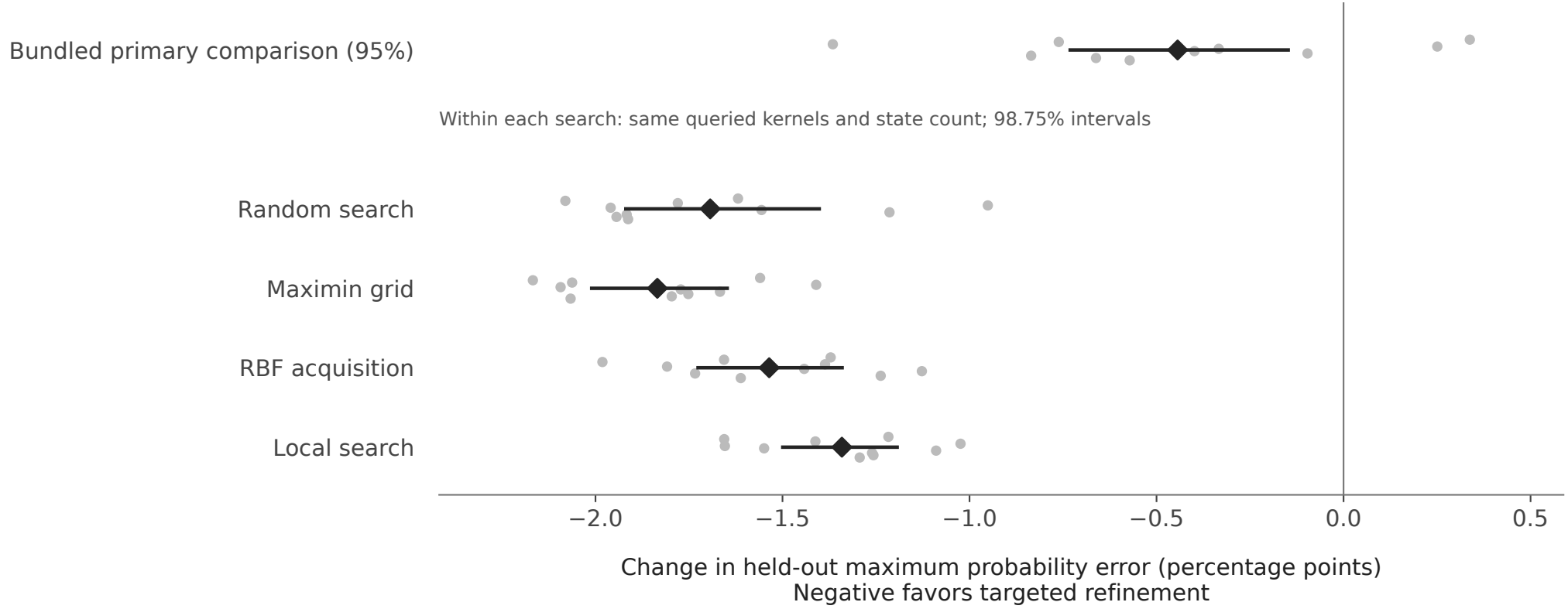
Fixed synthetic example: trust family, 128 states, seed 40000. RBF search, eight queries.
Witness environment (-1.000, 0.714); 32 original blocks, 122 refined blocks.



The changed kernel is known in this example. Forecast differences isolate abstraction error; they do not test environment learning.

Refinement versus matched-capacity controls

300 response-changing worlds; 16 fixed held-out environments; horizon 10.
Ten shared seed clusters, not individual worlds or predictions, determine uncertainty.

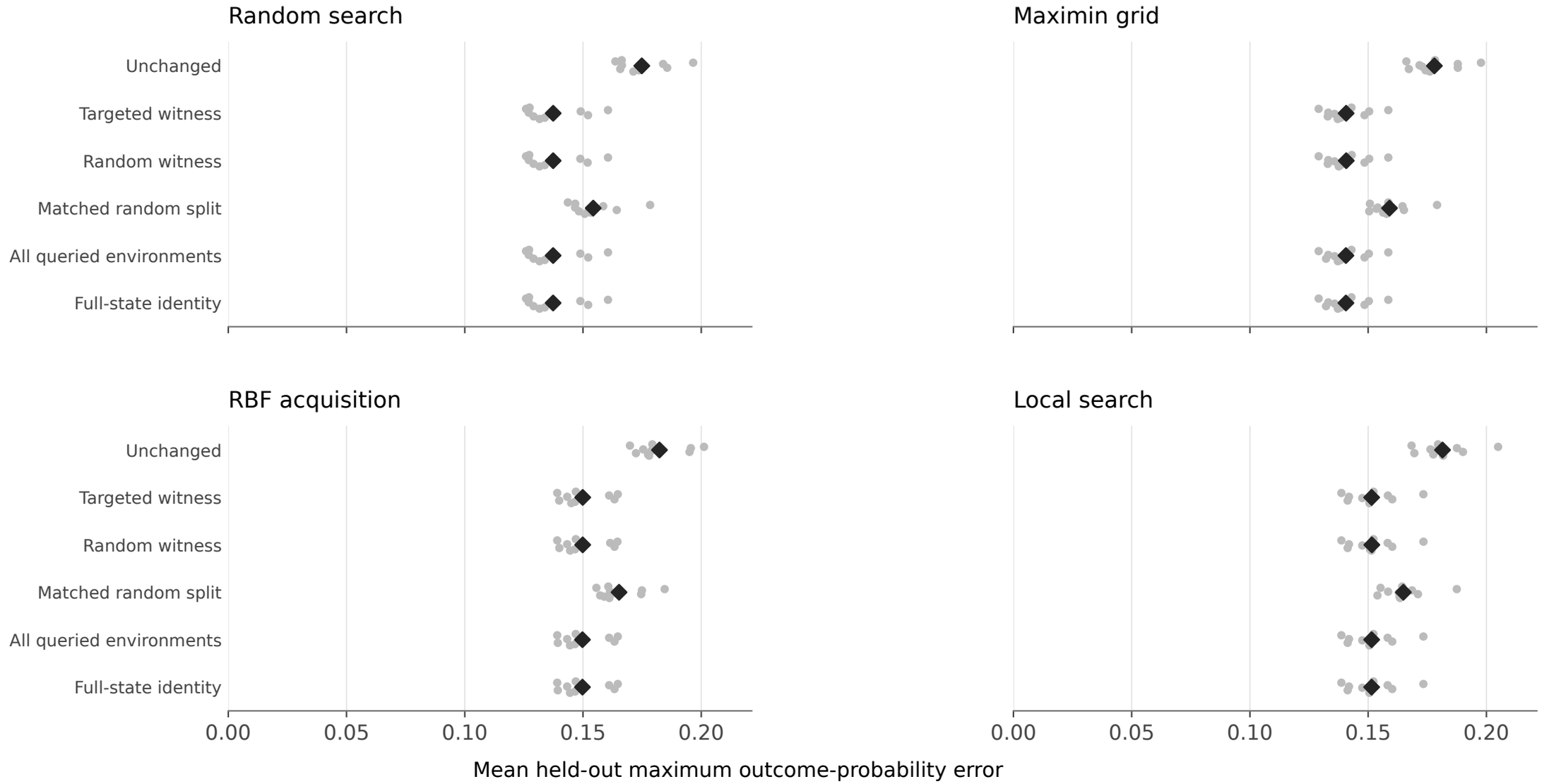


Gray dots: seed-level effects. Black diamond: paired mean. Lines: seed-cluster bootstrap intervals (5,000 replications).

Primary compares RBF/targeted with random-search/random-capacity strategies and can differ in capacity. The four lower rows isolate refinement.

All search methods and all partition controls

Descriptive horizon-10 frontier over all 300 response-changing worlds.
Every search has eight queries; only targeted and random-split controls within a method have exactly matched capacity.



Gray dots: ten seed-level means. Black diamonds: their mean. Full-state identity isolates response-surface learning error.